Exploring the Determinants of Docked Bike-Share Demands in Connection with Ferry Transit Integration

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Abstract: As urban environments evolve into increasingly complex mobility ecosystems, understanding multimodal integration is essential for sustainable urban planning. This study investigates integrated docked bike-sharing (DBS)–ferry demand in New York City using station-level data from summer 2024. Integrated trips were defined as potential transfers: Citi Bike trips ending or beginning within 12 minutes of a scheduled ferry departure or arrival at 16 ferry terminals. An XGBoost model was benchmarked against OLS, a Poisson GLM, and LightGBM using a temporal holdout. XGBoost and LightGBM performed almost identically ($R^2 = 0.85$) and clearly outperformed the linear models, so XGBoost was interpreted with SHapley Additive exPlanations (SHAP). Temporal and weather conditions accounted for 49.3% of total SHAP importance, followed by station characteristics (32.6%), socio-demographic attributes (9.1%), and built-environment characteristics (8.9%). Dock capacity and proximity to the ferry stop were the most influential station characteristics: larger stations and those closer to the ferry stop were associated with higher predicted integrated trips. Last-mile trips and weekends were also associated with higher predicted integrated trips. The negative contributions of the peak-period indicators reflect the unequal length of the aggregation periods, since demand per hour was highest during the peaks. The importance ranking was stable when the transfer window was varied from 8 to 20 minutes. These insights improve understanding of the under-explored integration of docked bike sharing and ferry transport, although evidence from other seasons and cities is needed to assess their generalizability.

Keywords: Active Mobility; Docked Bike-Sharing; Ferry transit; Integrated Trip; First and last-mile ; XGBoost; SHAP

1. Introduction

Urban areas worldwide are confronting a new era of mobility challenges, driven by mounting traffic congestion and a global imperative to reduce carbon emissions. In response, a paradigm shift is underway, moving away from car-centric models toward more sustainable, multimodal transportation systems (Hu, Xiong et al. 2021, Hu, Liao et al. 2025). This commitment to reshaping urban mobility is exemplified by landmark policy changes in major cities. For instance, NYC introduced North America's first congestion pricing initiative in January 2025, targeting vehicles entering Manhattan's central business district during peak hours. By imposing tolls, this policy aims to reduce traffic, improve air quality, and fund public transit upgrades. Similar approaches have been adopted in London and Stockholm to ease congestion and reduce traffic-related pollution (Santos and Shaffer 2004, Börjesson and Kristoffersson 2018).

This transformation is further fueled by innovative transportation solutions, such as bike-sharing services (BSS). These flexible solutions, including DBS, have become a cornerstone of modern urban planning by effectively solving the 'first and last-mile problem'—the critical gap between major transit hubs and a commuter's final destination (Jin, Cheng et al. 2025, Roig-Costa, Miralles-Guasch et al. 2025). The successful integration of new and traditional transport modes, a key strategy within the broader framework of multimodal transport, is central to creating more efficient, equitable, and environmentally friendly cities, especially in a dense and complex environment like NYC, where a well-established docked bike system complements its robust public transit network.

This mode integration, where DBS functions as a feeder to public transportation, has the potential to substantially enhance the efficiency and accessibility of the urban transit system (Yin, Rybarczyk et al. 2024, Banisharif-Dehkordi, Alesheikh et al. 2025, Sun, Cui et al. 2025). It can also mitigate transportation issues arising from automobile dependence, such as air pollution and traffic congestion. The operational success of a bike-sharing system, however, hinges on its ability to meet user demand, which is often characterized by notable spatiotemporal fluctuations. These fluctuations may result in scenarios where DBS stations are either full, preventing users from returning bikes, or stations are empty, depleting the supply for new users (Mellou and Jaillet 2019, Su, Yan et al. 2024). Such discrepancies underscore the necessity of comprehending spatial and temporal variations in integrated trips connected to public transit. A comprehensive understanding of DBS usage patterns in relation to public transit is essential for formulating effective policies and strategies to optimize bike-sharing operations and enhance overall urban mobility.

While the integration of BSS with public transit has been widely studied, the literature primarily focuses on its use at bus stations and metro systems (Huang, Yin et al. 2024, Javadiansr, Davatgari et al. 2024, Kosmidis and Müller-Eie 2024). However, the specific trips between DBS systems and ferry services remain a notably underexplored area in the current body of research. Given the growing importance of ferries in many coastal cities, there is a strong need for research examining the spatiotemporal relationships between these integrated trips and independent features near ferry terminals. A detailed analysis is particularly warranted, as the first- and last-mile segments of integrated journeys exhibit distinct characteristics that are heavily influenced by the trip's spatiotemporal context. For example, a morning commute to a commercial hub will have a different usage pattern than an evening trip returning to a residential area. A comprehensive understanding of these contextual factors is therefore essential for optimizing bike-sharing resource allocation, enhancing demand forecasting, and ultimately improving the user experience for those who depend on these multimodal connections, thereby decreasing reliance on motor vehicles.

Building on insights from previous studies, this research aims to delineate the factors influencing the integrated use of DBS and ferry transit, with particular emphasis on assessing the effects of temporal variables that fluctuate over time (e.g., precipitation, relative humidity) and spatial variables that are invariant over time (i.e., built environment, land use features) on this mode integration. Specifically, this study seeks to address the following two questions:

- Which variables influence integrated DBS–ferry demand, and what is the nature of their relationship with demand — linear, threshold, or otherwise non-linear ?
- How does first-mile versus last-mile contribute to integrated demand relative to other station-level and contextual factors ?

By achieving these objectives, this paper makes key contributions to the fields of transportation planning and urban mobility. Firstly, it offers a novel approach to understanding multimodal travel behavior by focusing on a previously underexplored transit combination. Secondly, the findings provide data-driven insights to inform policy decisions to optimize bike-sharing station placement near ferry terminals, improve service coordination, and enhance the overall efficiency of NYC’s multimodal transportation network. Lastly, because the workflow relies on openly available GBFS and GTFS data, it could be replicated in other cities with comparable bike-share and transit feeds, although the findings reported here are specific to New York City and summer 2024.

2. Literature Review

As the integration of DBS systems with ferry transit remains an underexplored area, this section reviews previous studies that examine how interactions and various factors affect BSS ridership when BSS serves as a transfer mode to or from transit stations.

2.1. Interactions between BSS and other modes of transportation

Research on the relationship between bike-sharing and public transit typically identifies three categories of interaction: integration, complementarity, and substitution (Kong, Jin et al. 2020, Li, Ha et al. 2024, Liu, Fang et al. 2024). These classifications enhance understanding of how bike-sharing fits within broader urban mobility patterns. Complementation occurs when BSS addresses gaps in public transit coverage, often seen with dockless systems that provide high availability and flexibility. Substitution refers to instances in which bike-sharing replaces a public transit trip, particularly during congested peak hours, when bikes can be a more efficient alternative in densely populated urban areas. The third category, integration, entails using bike-sharing to complete the initial and final segments of a journey that includes public transit (Kong, Jin et al. 2020, Politis, Fyrogenis et al. 2020).

The third category, integration, entails using bike-sharing to complete the initial and final segments of a journey that includes public transit (Kong, Jin et al. 2020, Politis, Fyrogenis et al. 2020). Accurately identifying and isolating these integrated trips is crucial for valid analysis.

A common technique for identifying integrated trips is to use a spatial buffer zone around a transit station. The radius of this buffer can vary, generally ranging from 30 to 300 meters, depending upon the mode of public transit (Javadiansr, Davatgari et al. 2024, Tushar and Buehler 2024, Zhu, Hu et al. 2024, Sun, Cui et al. 2025). Nevertheless, this methodology has inherent limitations, as a trip occurring within the buffer zone may not actually involve public transit. To improve accuracy, recent studies have adopted more sophisticated methods. For instance, Kong et al. (2020) incorporated public transit schedules to verify connections, whereas Li et al. (2024) applied a temporal threshold—based on the average time to travel two miles and the median trip duration—to discern short, integrated trips, considering that the duration of the bike-share segment was anticipated to be short.

The existing literature on BSS integrated patterns has primarily focused on understanding general usage trends, with temporal analyses highlighting variations by season, day of the week, and, importantly, peak hours (Zhu, Hu et al. 2024, Banisharif-Dehkordi, Alesheikh et al. 2025), which provides a critical foundation for rebalancing strategies. Therefore, our research builds upon this established understanding of temporal patterns by examining how DBS are utilized as

a first-mile (access) or last-mile (egress) solution, specifically during morning and evening peak commutes.

2.2. Determinants of Bike-Sharing and Public Transit Integration

Numerous studies have examined how various factors influence the integration of bike-sharing systems with public transit modes, such as metro rail and bus services. However, there is a scarcity of research that concurrently considers integrated trips involving DBS and ferry transit. Among the many factors, identifying those that most significantly influence this relationship is essential to understanding the mechanism of interaction between BSS and public transit systems. Consistent with the empirical framework developed in this study, the variables most frequently highlighted by previous studies are reviewed below under four categories: (bike) station characteristics, built environment, socio-demographic characteristics, and temporal factors.

Station characteristics variables describe the physical and operational capacity of the bike-sharing system itself. For DBS systems specifically, the number of available docks at a station is generally positively correlated with usage, since sufficient dock availability reduces the risk of failed pickups or drop-offs that would otherwise deter riders from choosing BSS for a transit connection (Ntamwiza and Bwire 2025). Proximity between a station and the transit facility it serves — typically operationalized as network or Euclidean distance — has similarly been identified as a strong determinant of integration, reflecting a standard distance-decay relationship in first- and last-mile travel behavior (Wei and Zhu 2023, Lu, Li et al. 2026)

Previous studies have demonstrated that the built environment around public transit stations significantly influences the use of feeder BSS (Guo, Yang et al. 2022, Gou, Feng et al. 2025, Hu, Liao et al. 2025). Several studies highlight the influence of land-use features — including the distribution of different land types, the location and density of activities, and proximity to public transit facilities. For example, Zuniga-Garcia et al. (2022) found that commuting trips on university campuses and in downtown Austin are largely driven by land use. Similarly, Duran-Rodas et al. (2019) studied built environment factors affecting bikeshare ridership in six German cities. They found that population density, distance from the city center, and distance to amenities (like bakeries) were important variables. In general, density-based factors, such as road, residential, and population density, are typically positively correlated with integrated usage (Guo, Yang et al. 2022, Yang, Jin et al. 2026). However, findings on density can be complex and sometimes contradictory: for instance, Zhou et al. (2023) found that higher density of public transit stops negatively affects the integrated use of bikes and public transit, whereas land use positively influences bike-transit integration.

Socio-demographic attributes of riders shape how individuals interact with transportation systems, reflecting broader patterns of income, ethnic identity, education, age, marital status, and physical health in relation to mobility (Ren 2025), though findings can be complex and sometimes contradictory. Gender is a consistent factor, with most studies observing that male individuals are more likely than women to combine BSS with public transit (Böcker, Anderson et al. 2020). Age also plays an important, but contested, role. Some studies argue that older individuals are less likely to use BSS as a mode of first- or last-mile connectivity (Liu, Liu et al. 2026, Yang, Zhu et al. 2026). Findings on the role of income are similarly contradictory. Some studies indicate a positive association between higher income and the use of shared bikes combined with public transit (Ji, Fan et al. 2017), while others report a negative impact (Guo, Bhat et al. 2007, Yang and Wang 2014) or find that median income groups are the most likely to be users (Zhao, Yuan et al. 2022, Mohiuddin, Fitch-Polse et al. 2024, Zhang, Zheng et al. 2025). Despite these tensions, research consistently shows that higher educational attainment correlates with a greater openness to cycling for transportation (Jonkeren, Kager et al. 2021, Mitra and Hess 2021). Beyond individual demographics, other social and household factors influence modal choice. Some studies have found that ethnic minority groups are less likely to cycle than other groups (Moudon, Lee et al. 2005). Furthermore, household size and per-capita vehicle access also influence modal choice, with larger families and higher per-person car ownership favoring driving for commutes (Pucher and Buehler 2008, Zhou, Huang et al. 2025). Additionally, accessibility to employment opportunities is a key factor, as some research indicates lower BSS usage in disadvantaged communities due to a lack of nearby jobs (Su, Yan et al. 2024, Zhang, Zheng et al. 2025).

Temporal factors complete the four-category framework and are critical to understanding when, rather than just where, integration occurs. Time of day and day of week are particularly influential: BSS–transit integration is typically strongest on weekdays, consistent with commute-driven travel, and may shift toward a substitution relationship on weekends when trips are more discretionary or recreational (Kong, Jin et al. 2020, Zhu, Wang et al. 2026). Research also confirms that BSS usage is highly sensitive to weather, with warm and dry weather increasing demand, while extreme weather significantly reduces it (Cantelmo, Kucharski et al. 2020, Jaber, Csonka et al. 2022, Guzel, Altintasi et al. 2025).

Taken together, previous studies identify station characteristics and, to a lesser extent, temporal factors as important determinants of BSS–transit integration, while findings on the built environment and socio-demographic characteristics remain mixed and context-dependent. However, recent studies have focused predominantly on metro- and bus-based integration, leaving ferry-based integration largely unexplored. The present study addresses this gap directly,

applying the same four-category framework to a DBS–ferry context and evaluating whether the determinants and their relative importance identified in the metro/bus literature generalize to a fundamentally different transit mode.

3. Materials and Methods

The research framework, depicted in Figure 1, presents the overall methodology, which comprises four primary stages: (1) trip data collection and preprocessing, (2) feature engineering and selection, (3) machine learning model development and optimization, and (4) findings.

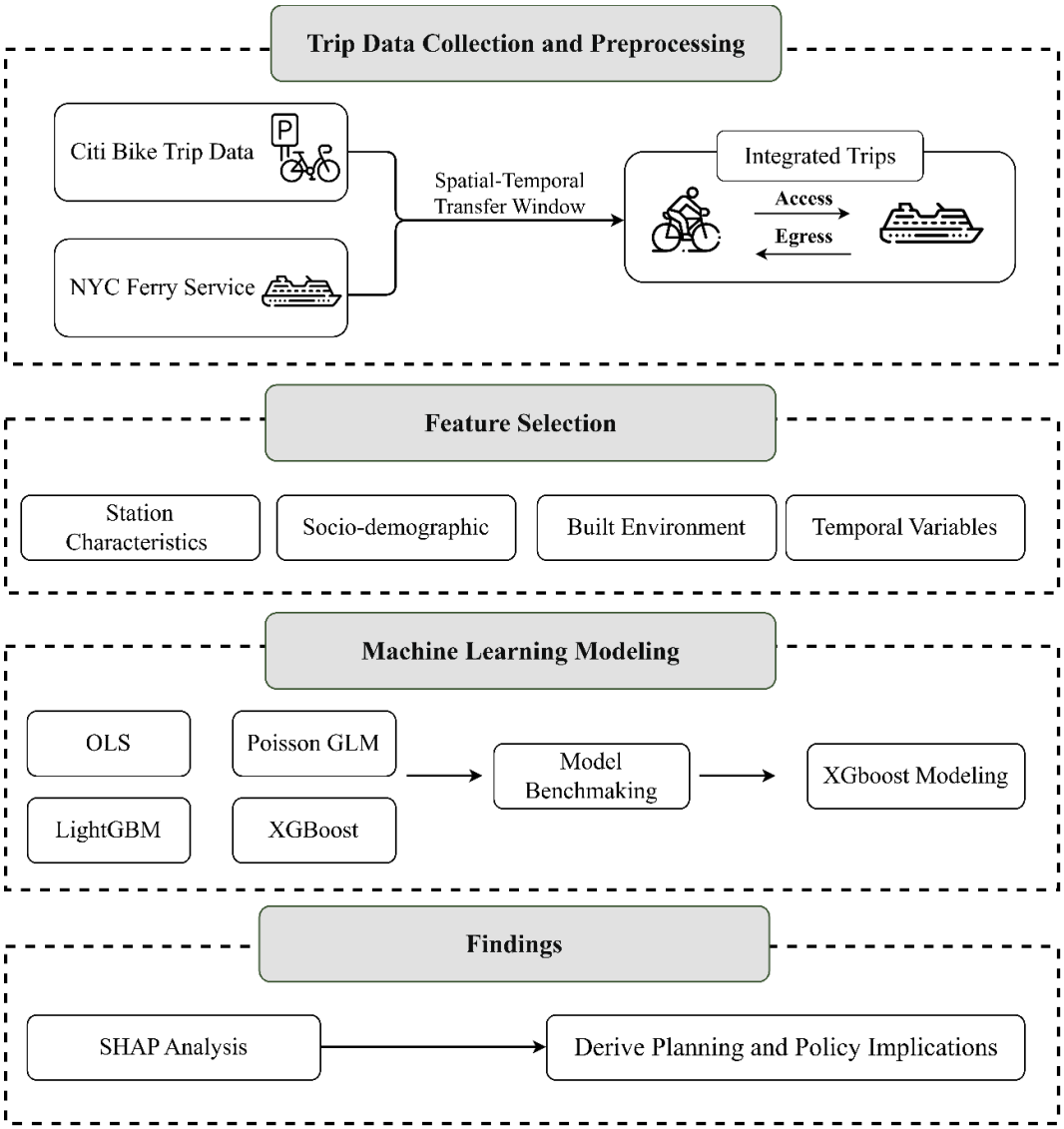


Fig. 1. Methodology Framework

3.1. Study area and data

This study investigates integrated DBS trips connected to the NYC Ferry system. The analysis combines two primary mobility datasets: Citi Bike trip records and the NYC Ferry schedule in the General Transit Feed Specification (GTFS) format. These datasets are complemented by built-environment, socio-demographic, and weather data to capture the factors influencing integrated bike–ferry travel demand. The study focuses on the summer months of 2024 (June–August), when both Citi Bike and NYC Ferry experience their highest ridership levels (Citi Bike 2024, NYC Ferry 2024).

The Citi Bike dataset contains trip-level information, including origin and destination stations, departure and arrival times, trip duration, and user type. During the study period, approximately 14 million trips were recorded across 2,341 docking stations. The NYC Ferry system, which recorded over 7.1 million annual passenger boardings in 2024, operates 25 ferry terminals throughout the city's five boroughs. GTFS data were used to obtain ferry routes, terminal locations, and service schedules needed to identify integrated trips.

Because integrated bike–ferry trips can occur only where both transportation systems are spatially connected, ferry terminals lacking a Citi Bike station within a 300-meter buffer were excluded from the analysis. These exclusions (comprising nine terminals) primarily reflect outer-borough landings where the Citi Bike service network had not yet expanded during the study period. Figure 2 shows the final study area, which consists of 16 fully integrated ferry terminals and their associated Citi Bike stations. Furthermore, Figure 3 illustrates the hourly distribution of average bike-sharing demand on weekdays and weekends, highlighting distinct morning (07:00–10:00) and evening (16:00–19:00) peak hours.

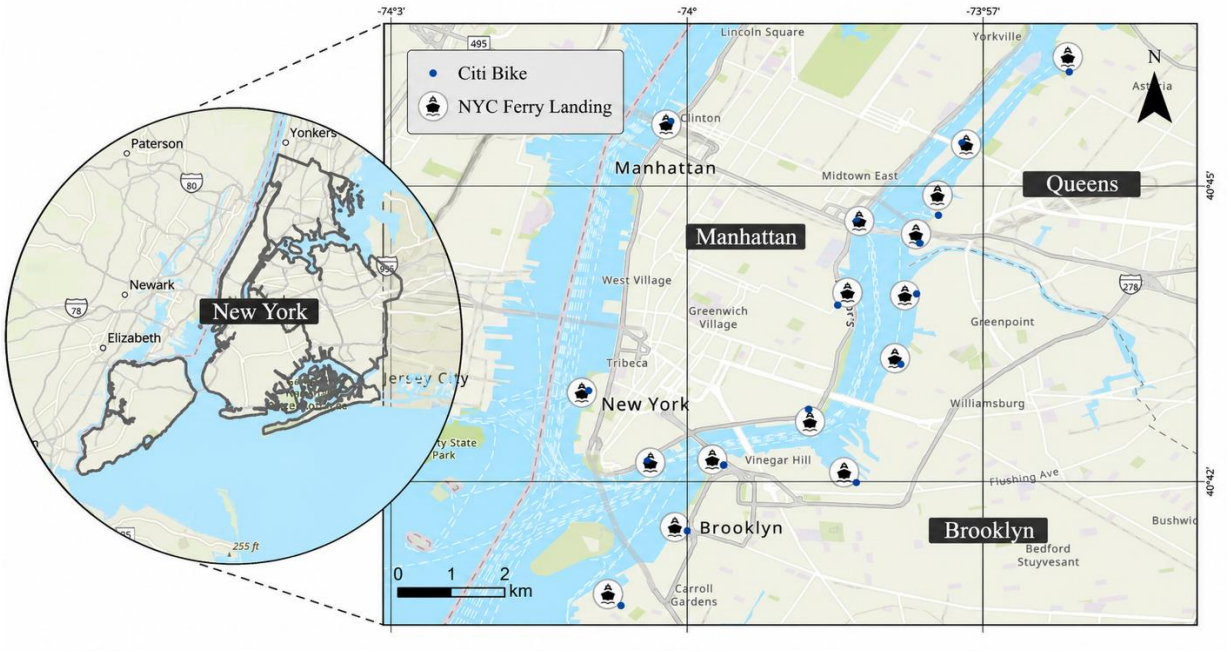


Fig. 2. Study area and dataset

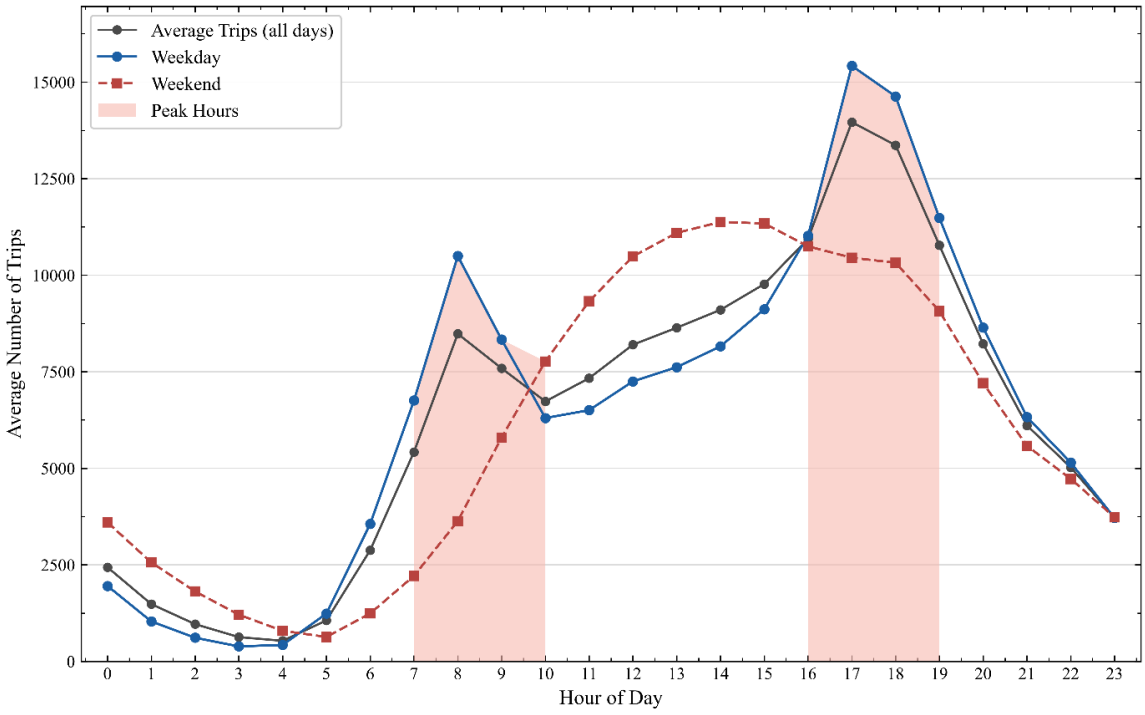


Fig 3. Average hourly system-wide citibike trip demand (June–August 2024)

3.2. Identifying Integrated Trips

Preprocessing of the raw Citi Bike trip data was performed to ensure a more robust and representative analytical dataset. Trips with durations shorter than 2 minutes were removed to eliminate accidental dockings and potential system errors, while trips exceeding 90 minutes were excluded to filter out leisure rides and bicycle maintenance events.

Because Citi Bike data do not explicitly indicate trip purpose or modal transfers, integrated ferry–bike trips must be inferred using a data-driven approach. Recent studies have proposed spatiotemporal methods for reconstructing integrated trips from shared mobility datasets (Li, Du et al. 2020, Guo and He 2021, Wu, Lu et al. 2021, Li, Ha et al. 2024, Lu, Li et al. 2026). Many previous studies rely primarily on spatial proximity or buffer-based methods, with transit-stop buffers ranging from 50 to 500 m (Wu, Lu et al. 2021, Cheng, Wang et al. 2022, Javadiansr, Davatgari et al. 2024, Zhu, Hu et al. 2024). This study links Citi Bike stations to ferry terminals using a 300 m buffer around the ferry stop. At some terminals the distance between the ferry stop, where ferry trips start and end, and the terminal entrance exceeds 200 m, so a smaller buffer would miss stations that are close to the entrance but far from the stop. In the dense urban environment of NYC, however, numerous Citi Bike stations lie within walking distance of ferry terminals, so a spatial criterion alone would produce many false-positive matches, and a more restrictive temporal criterion is required.

Accordingly, this study defines an integrated trip as a Citi Bike trip that ends at a linked station within 12 minutes before a scheduled ferry departure (access trip) or begins within 12 minutes after a scheduled ferry arrival (egress trip) at the nearest ferry terminal. Because the data record only the temporal association between bike trips and ferry service, matched trips are potential rather than confirmed transfers. This temporal threshold, adapted from previous studies (Banisharif-Dehkordi, Alesheikh et al. 2025, Ren 2025), accounts for walking time between stations (located up to 300 meters from their terminals) and terminal platforms, passenger boarding and alighting processes, and transfer delays at high-demand terminals. To assess the influence of this parameter, transfer windows ranging from 8 to 20 minutes were tested (Figure 4). The number of identified trips increased steadily with window length, from 177,530 at 8 minutes to 310,278 at 20 minutes. While wider windows cover a larger portion of the ferry timetable and therefore increase the chance of ambiguous or coincidental matches, shorter windows leave insufficient time for commuters needing to walk between the station and the terminal, while providing minimal margin for boarding. The 12-minute window was retained as an intermediate choice, identifying 231,902 trip records. A sensitivity analysis of the feature-importance ranking to the transfer window is provided in Appendix A. The conceptual framework for identifying access (first-mile) and egress (last-mile) trips is illustrated in Figure 5.

Figure 6 shows the average hourly profile of the identified integrated trips on weekdays and weekends. At the macro level, it follows the same temporal pattern as system-wide Citi Bike demand in Figure 3. Weekday trips show distinct morning and evening peaks, with the evening peak being higher, while weekend trips rise gradually to a broad midday–afternoon plateau. This

agreement suggests that the identified integrated trips reflect the general travel rhythm of the bike-share system. The absence of integrated trips between approximately 23:00 and 05:00 reflects the lack of ferry service, not low DBS demand.

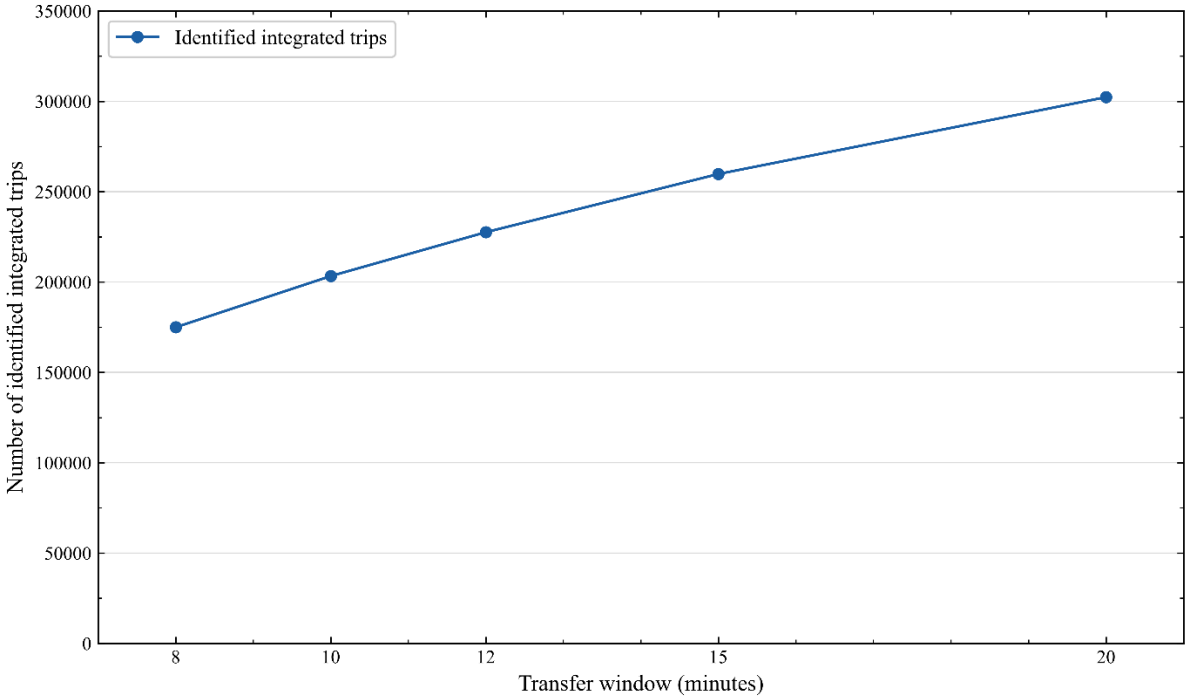


Fig 4. Identified integrated bike–ferry trips to alternative transfer window thresholds

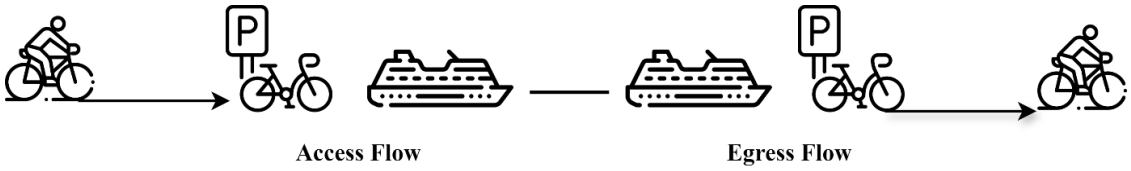


Fig 5. Distinct access (first-mile) and egress (last-mile) components of integrated bike–ferry trips.

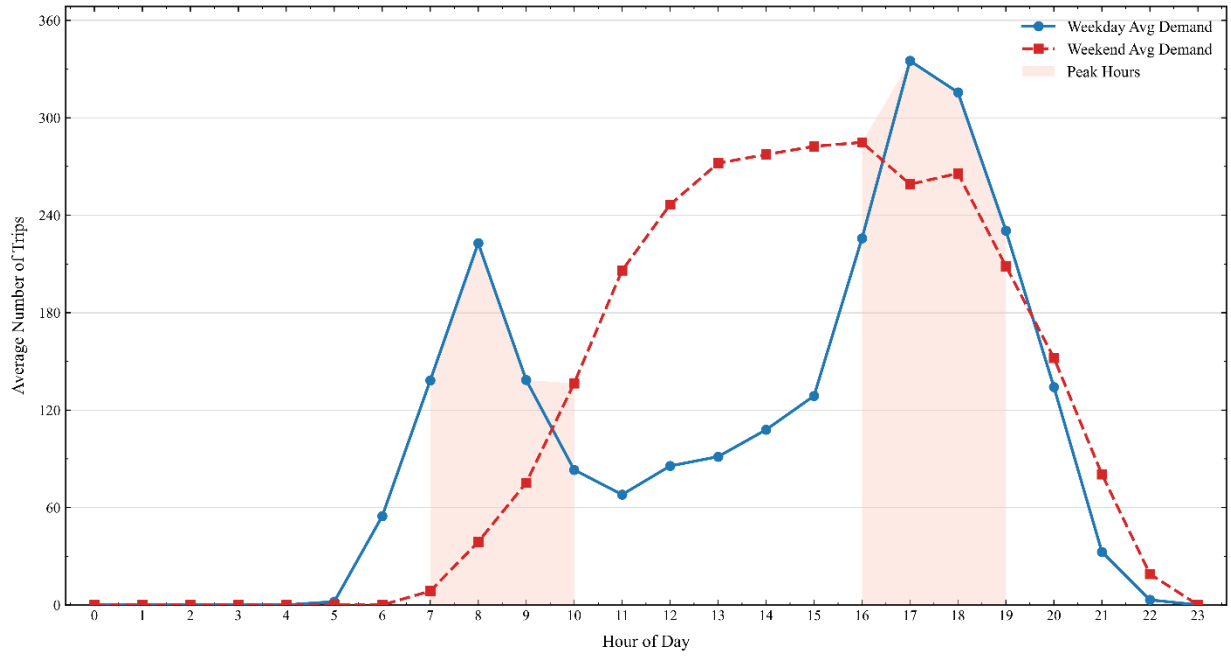


Fig 6. Hourly distribution of integrated DBS–ferry trips using a 12-minute threshold

3.3. Deriving Independent Variables

To identify the factors that drive integrated bike–ferry demand, candidate explanatory variables were compiled from multiple spatial and temporal datasets, guided by existing studies on bike-sharing demand and multimodal accessibility. Spatial variables were evaluated within a 500 m catchment around each bike station, hereafter referred to as the catchment. The variables were organized into four categories: station characteristics, built-environment, socio-demographic, and temporal and weather variables (Table 1).

Station characteristics comprised dock capacity, ferry distance, and trip stage. Dock capacity was obtained from the General Bikeshare Feed Specification (GBFS) station information. Ferry distance was computed as the straight-line (Euclidean) distance between each bike station and its designated ferry terminal landing, using GBFS station coordinates and GTFS ferry terminal geolocations. A binary indicator distinguished first-mile access trips to ferry terminals from last-mile egress trips

Socio-demographic variables were compiled from the U.S. Census Bureau's 5-Year American Community Survey (ACS 2024) and the Longitudinal Employer-Household Dynamics Origin–Destination Employment Statistics (LEHD LODES) Workplace Area Characteristics file (U.S. Census Bureau 2023). ACS measures were assigned to each station from the census tract containing it, and included population density, age composition (shares of residents aged 18–24 and 25–30), educational attainment (share of residents with a bachelor's degree or higher), median household income, and the share of zero-vehicle households as an indicator of automobile

availability. Employment density from LODES represented local employment activity around each station.

Built-environment variables captured the immediate micro-environment of each station. Land-use composition (residential, commercial, industrial, and institutional shares) was derived within the catchment from the NYC Department of City Planning MapPLUTO tax-lot database (NYC Department of City Planning 2024), and land-use diversity was summarized with a normalized land-use entropy index. Total road length within the catchment was extracted from OpenStreetMap (OSM). Intersection density and transit accessibility were taken from the U.S. Environmental Protection Agency Smart Location Database (SLD), which reports urban-form and transit-supply measures at the census block-group level (U.S. Environmental Protection Agency 2021). To assign these block-group values to stations, the catchment was intersected with the block-group polygons, each value was multiplied by the share of the block group's area falling within the catchment, and the weighted values were summed over all block groups overlapping the catchment.

Finally, temporal and weather variables were included to account for short-term demand fluctuations. Daily maximum temperature, precipitation, maximum wind speed, and mean relative humidity were obtained from the National Oceanic and Atmospheric Administration (NOAA) weather station network (NOAA 2024). The temporal structure of demand was represented by indicator variables for the morning peak (07:00–10:00) and evening peak (16:00–19:00), with off-peak as the reference, and for weekend versus weekday. The dependent variable was the number of integrated trips per bike station, day, period (morning peak, evening peak, or off-peak), and trip stage (first mile or last mile). Combinations with no integrated trips were set to zero, giving 8,832 observations (16 stations × 92 days × 3 periods × 2 trip stages), of which the final 14 days (1,344 observations) formed the test set. The off-peak period covers the remaining 18 hours; because ferry services do not operate overnight (approximately 23:00–05:00, with terminal-specific schedules), it contains roughly 12 hours of effective service. Because the periods differ in length, a sensitivity model based on hourly rates is reported in Appendix B.

Table 1. Descriptive statistics and data sources of the explanatory variables

Feature Category	Variable	Description	Max	Mean	St. Dev.	Data Source
Station characteristics	Dock capacity	Number of docking points at the anchor bike station	123.00	44.68	24.38	Citi Bike System Data
	Ferry distance (m)	Euclidean distance between the bike station and its connected ferry terminal	297.11	181.50	86.88	Citi Bike / NYC Ferry GTFS
	Trip_stage	Binary flag for trip stage (1 = First-Mile, 0 = Last-Mile)	1.00	0.50	0.50	Citi Bike Trip Data
Built environment	Residential (%)	Share of residential land use within the 500 m buffer	38.19	18.82	8.77	NYC MapPLUTO
	Commercial (%)	Share of commercial land use within the 500 m buffer	18.89	4.49	5.05	NYC MapPLUTO
	Industrial (%)	Share of industrial land use within the 500 m buffer	58.67	11.49	13.86	NYC MapPLUTO
	Institutional (%)	Share of institutional land use within the 500 m buffer	12.83	2.85	3.05	NYC MapPLUTO
	Land-use entropy	Normalized entropy index measuring land-use mix diversity within the 500 m buffer	0.93	0.72	0.12	NYC MapPLUTO
	Intersection density	Number of street intersections per unit area within the 500 m buffer	1818.80	704.08	546.09	SLD
	Transit access	Composite measure of proximity and access to non-ferry public transit services	6.81	1.83	1.80	SLD
	Road length (m)	Total street network length within the station's 500 m buffer	76659.79	59949.28	12518.77	OSM
Temporal variables	Is weekend	Binary indicator (1 = Saturday/Sunday, 0 = weekday)	1.00	0.29	0.45	Calendar / Citi Bike trip data
	Morning peak time	Binary flag for AM peak hours (07:00–10:00)	1.00	0.33	0.47	Citi Bike Trip Data
	Evening peak time	Binary flag for PM peak hours (16:00–19:00)	1.00	0.33	0.47	Citi Bike Trip Data
	Max temperature (°C)	Daily maximum air temperature	35.50	29.16	3.22	NOAA

Table 1. (continued)

Feature Category	Variable	Description	Max	Mean	St. Dev.	Data Source
Temporal variables	Precipitation (mm)	Daily total precipitation	35.40	4.75	8.91	NOAA
	Max wind speed (km/h)	Daily maximum sustained wind speed	43.38	19.80	5.62	NOAA
	Mean humidity (%)	Daily mean relative humidity	89.38	69.60	10.71	NOAA
Socio-demographic	Median income ($)	Area-weighted median household income of the catchment tract(s)	159219.28	109059.84	39126.18	ACS
	Education (%)	Share of population aged 25+ holding a bachelor's degree or higher	71.10	52.45	14.91	ACS
	Employment density	Weighted count of jobs (workplace employment) within the station's catchment area	47195.33	8275.36	11890.15	LEHD LODES
	Zero-vehicle households (%)	Share of households with no vehicle available	54.33	31.63	10.55	ACS
	Age 18–24 (%)	Share of population aged 18–24	16.63	5.96	3.62	ACS
	Age 25–30 (%)	Share of population aged 25–30	47.46	25.84	11.07	ACS
	Population density	Residential population per km² within the station's catchment area	28023.81	14424.01	5823.70	ACS

3.4. Extreme gradient boosting

XGBoost was adopted as the primary machine learning algorithm to model integrated bike-sharing demand throughout the study period. XGBoost has demonstrated strong predictive performance in transportation research by capturing complex nonlinear relationships and interactions among explanatory variables (Zhu, Hu et al. 2024, Shen, Weng et al. 2025, Yang, Zhu et al. 2026). Unlike conventional regression approaches, XGBoost requires no prior assumption about the functional form of the relationship between predictors and the response variable, and it is comparatively robust to multicollinearity, missing values, and outliers — properties that make it well suited to heterogeneous, secondary-source urban datasets such as the one assembled here. The XGBoost prediction for observation (i) is expressed as:

$$\hat{y}i = \sum_{k=1}^{k} f_k(x_i)\,, \qquad f_k \in F$$

where ($\widehat{y_i}$) is the predicted value, (f_k) denotes the prediction from (k)-th decision tree, and (F) represents the space of all possible regression trees.

The model is trained by minimizing an objective function combining a loss term and a regularization term, which together improve generalization and reduce overfitting. At boosting iteration t, the objective is:

$$Obj^{(t)} = \sum_{i=1}^{n} L\left(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i)\right) + \Omega(f_t)$$

where $L(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i))$ is the loss function that measures the prediction error, ($\widehat{y_i^{(t-1)}}$) is the prediction from the previous iteration, ($f_t(x_i)$) is the newly added tree, and ($\Omega(f_t)$) is the regularization term that penalizes model complexity. This regularization enables XGBoost to achieve high predictive accuracy while maintaining strong generalization performance.

The predictive performance of XGBoost depends strongly on its hyperparameters, which were therefore tuned systematically. To respect the temporal ordering of the data, tuning used random search with expanding-window time-series cross-validation: the training period was divided into three consecutive folds in which each validation fold follows its training data, so that no future observations inform the fit. 30 candidate configurations were sampled from the search space (number of trees 300–700, maximum depth 4–8, learning rate 0.03–0.1, row subsampling 0.7–1.0, column subsampling 0.6–1.0, and L1 and L2 regularization), and the configuration with the lowest cross-validated RMSE was selected. The selected hyperparameters were 300 boosting trees, a maximum tree depth of 8, a learning rate of 0.03, a row subsampling ratio of 0.70, a

column subsampling ratio of 0.60, an L1 regularization coefficient (α) of 0.1, and an L2 regularization coefficient (λ) of 10.0. The model was then trained on data before August 18, 2024, and evaluated on the final 14 days of the study period, which were withheld from both training and tuning. The fitted model was used for prediction and SHAP-based interpretation.

3.5. Interpretation Methods

Although XGBoost provides high predictive accuracy, its ensemble structure limits direct interpretation of the relationships between predictors and the response variable. Therefore, to improve model transparency and quantify the contribution of each explanatory variable to integrated bike-sharing demand, SHapley Additive exPlanations (SHAP) were employed. SHAP enables both global and local interpretation of the trained model by estimating the marginal contribution of each feature to individual predictions, thereby providing a comprehensive understanding of the nonlinear effects and interactions learned by the XGBoost model. SHAP values quantify each feature's contribution to the model's output: positive values indicate features that increase predicted demand, while negative values indicate features that decrease it. SHAP values were computed for all observations, and global importance was defined as the mean absolute SHAP value of a feature.

3.6. Model Evaluations

The predictive performance of the proposed machine learning models was assessed using three widely accepted evaluation metrics: the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). R^2 measures the proportion of variance in the observed data explained by the model, whereas MAE and RMSE quantify the average magnitude of prediction errors; RMSE assigns greater weight to larger errors due to its squared residual term, making it particularly sensitive to substantial prediction deviations. In general, higher R^2 and lower MAE and RMSE values indicate better predictive performance. The metrics are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where y_i is the observed integrated demand for station-observation i, $\hat{y}_i$ is the predicted demand, $\bar{y}$ is the mean of observed values, and n represents the total number of evaluation samples. To guard against temporal data leakage and provide a realistic assessment of out-of-time prediction accuracy, model evaluation used a temporal holdout scheme: models were trained on the primary portion of the summer 2024 study period and evaluated on the final two weeks, ensuring that no observation in the evaluation set was seen during training.

All computational modeling and statistical analyses were performed using Python 3.11, and spatial visualizations were generated using ArcGIS Pro 3.4.

4. Results

4.1. Model Performance

The predictive performance of the proposed XGBoost model was benchmarked against baseline and alternative machine learning algorithms, including Ordinary Least Squares (OLS), a Poisson Generalized Linear Model (Poisson GLM), and LightGBM, using an identical temporal holdout validation scheme. Models were trained on the primary summer period and evaluated on the final two weeks of the study period (18 to 31 August 2024), which ensures a strict assessment of out-of-time generalization and prevents temporal leakage. Table 2 summarizes the out-of-time prediction accuracy of each algorithm.

Table 2. Comparative analysis of model result accuracy.

Model	RMSE	MAE	R^2
OLS	18.32	12.12	0.59
Poisson GLM	16.48	10.27	0.66
XGBoost	10.98	6.79	0.85
LightGBM	10.94	6.84	0.85

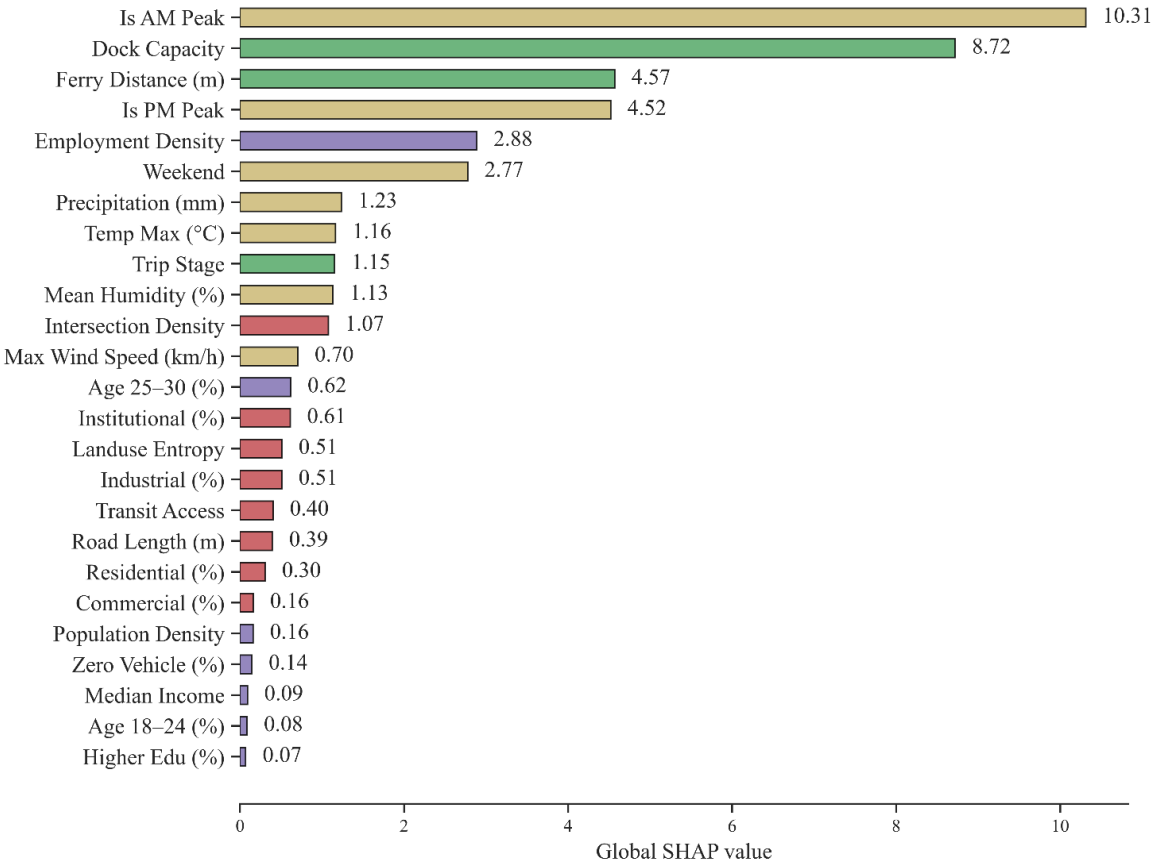
Both XGBoost and LightGBM substantially outperformed the linear models, which demonstrates the advantage of nonlinear approaches for predicting integrated travel demand. The two gradient-boosting models achieved very similar performance, with R^2 values of 0.85 and RMSE values of 10.98 and 10.94 for XGBoost and LightGBM, respectively. Given their comparable performance, XGBoost was selected for the subsequent SHAP-based analysis, considering its widespread use in similar studies (Zhu, Hu et al. 2024, Wang, Zhang et al. 2025, Lu, Li et al. 2026, Pham, Ngo et al. 2026, Zhu, Wang et al. 2026).

4.2. Variable importance

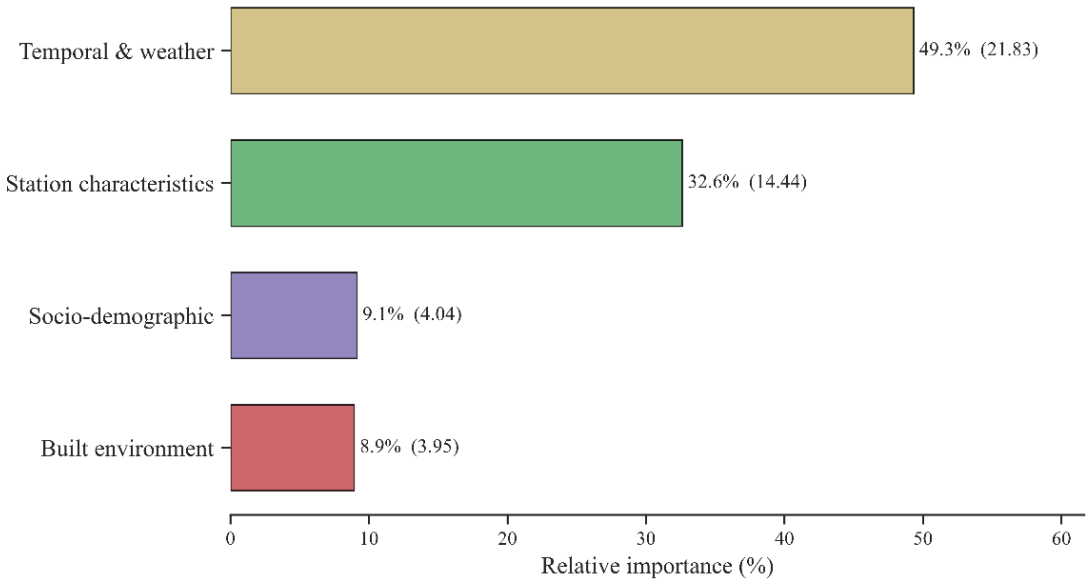
Figure 7 presents the global SHAP importance of the individual explanatory variables (Fig. 7a) alongside the aggregated importance of the four feature categories (Fig. 7b). Global importance is the mean absolute SHAP value of a variable across all observations, so it measures how strongly a variable shifts the model's predictions in either direction, not whether the shift is positive or negative. The direction and shape of each relationship are examined in the next section.

At the category level, temporal and weather conditions, which encompass the peak-period indicators, the weekend indicator, and meteorological factors, accounted for the largest share of predictive importance (49.3%). They were followed by station characteristics (32.6%), which encompass dock capacity, ferry distance, and trip stage. Socio-demographic attributes (9.1%) and built-environment characteristics (8.9%) played secondary roles, and their shares were very similar. Together, temporal and weather conditions and station characteristics accounted for 81.9% of the total importance. Predicted integrated demand is therefore associated mainly with when a trip takes place, the weather on that day, the capacity of the station, and its distance from the ferry stop, while the characteristics of the surrounding catchment add comparatively little once these factors are accounted for.

At the individual-feature level (Fig. 7a), the morning peak indicator (Is AM Peak) had the highest mean absolute SHAP value (10.31), followed by Dock Capacity (8.72), Ferry Distance (4.57), and the evening peak indicator (Is PM Peak, 4.52). Employment Density (2.88) and Weekend (2.77) formed the next tier. Precipitation (1.23), Maximum Temperature (1.16), Trip Stage (1.15), Mean Humidity (1.13), and Intersection Density (1.07, the most important built-environment variable) each had a value of about 1, followed by Maximum Wind Speed (0.70). Within the socio-demographic group, the share of residents aged 25–30 (0.62) was far more important than the share aged 18–24 (0.08), indicating that catchments with a larger share of residents aged 25–30 are associated with higher predicted demand. The remaining socio-demographic variables, Population Density (0.16), Zero-Vehicle Share (0.14), Median Income (0.09), and Higher Education (0.07), ranked at the bottom of the predictor set. Institutional share (0.61), Land-use Entropy (0.51), and Industrial share (0.51) were the next most important built-environment variables. The importance of the two peak indicators partly reflects the unequal length of the aggregated periods and not only commuting behavior, as explained in Section 4.3.



(a)



(b)

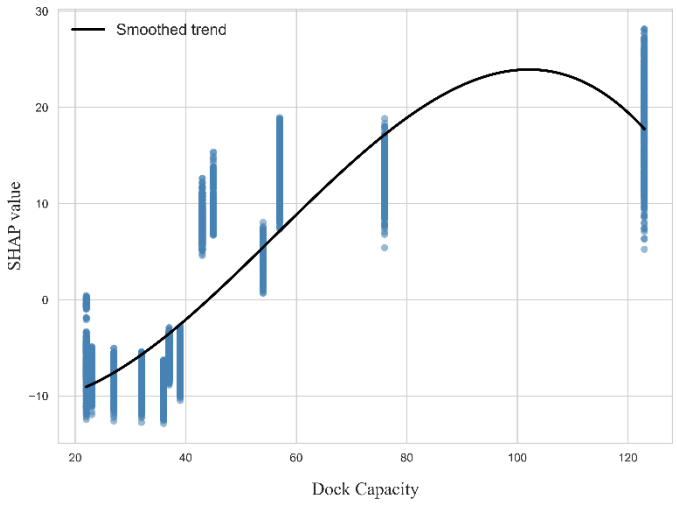
Fig. 7. SHAP-based global importance of explanatory variables (a) and relative importance of feature categories (b).

4.3. Determinant Plots

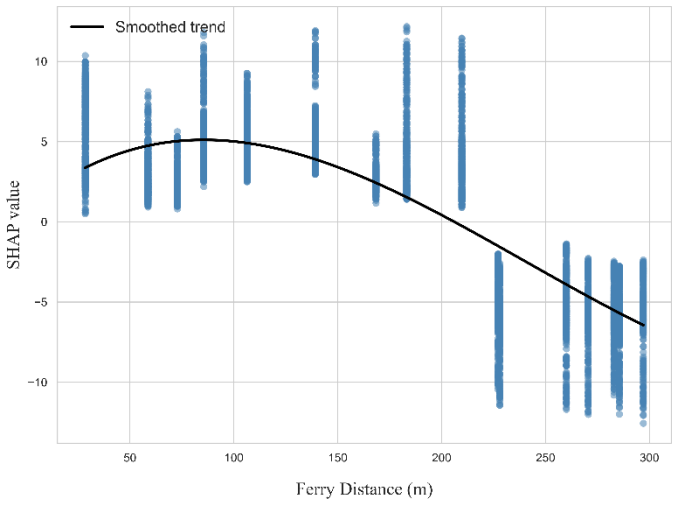
While the global importance analysis shows the overall ranking of variables, it does not show how each variable relates to predicted integrated DBS–ferry demand. SHAP dependence plots were therefore used to examine the contribution of each variable (Figs. 8–11). In these plots, points above the dashed zero line indicate a positive contribution to predicted integrated demand, and points below it indicate a negative contribution. The black line is a smoothed trend. Because station-level variables take one value per station, the smoothed line interpolates across gaps between stations, so patterns are described by the observed station values and the reported thresholds are approximate.

4.3.1. Station characteristics

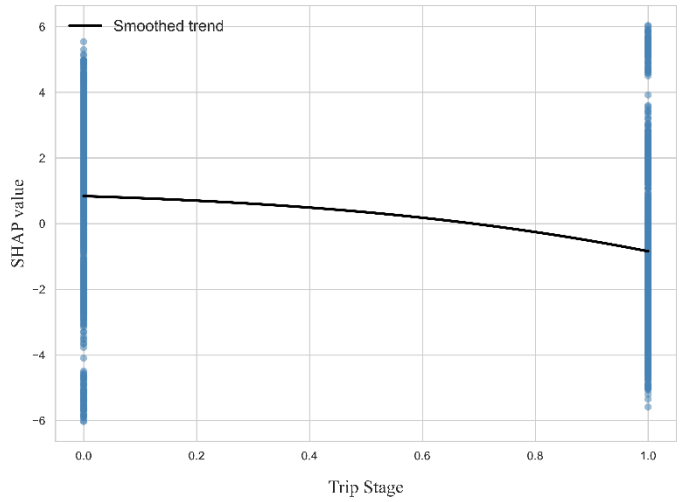
Figure 8 illustrates the contributions of station characteristics. Dock capacity (Fig. 8a) exhibits a pronounced nonlinear relationship: its contribution is negative for stations with fewer than approximately 40 docks, becomes positive at around 45 docks, and reaches its highest values at stations with 76 and 123 docks. Ferry distance (Fig. 8b) shows a threshold-like pattern, with stations located within approximately 210 m of the ferry stop contributing positively, while those located more than approximately 225 m away contribute negatively. Finally, trip stage (Fig. 8c) shows a modest difference, with last-mile trips (Trip Stage = 0) having a slightly more positive average contribution than first-mile trips (Trip Stage = 1), although the two ranges overlap substantially. This indicates somewhat higher predicted demand for last-mile than for first-mile trips.



(a)



(b)

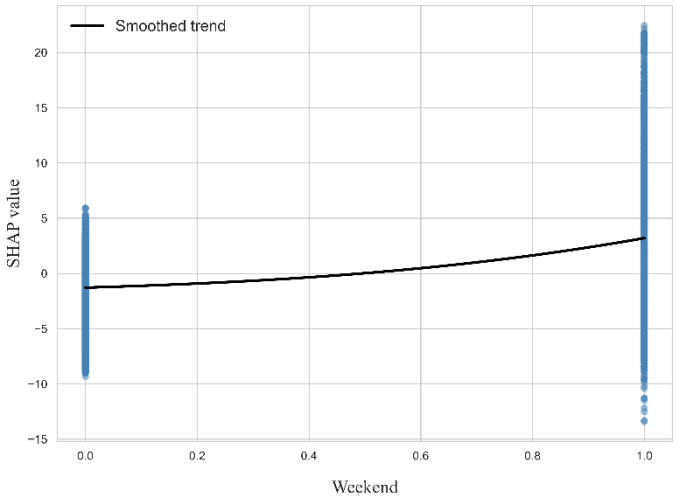


(c)

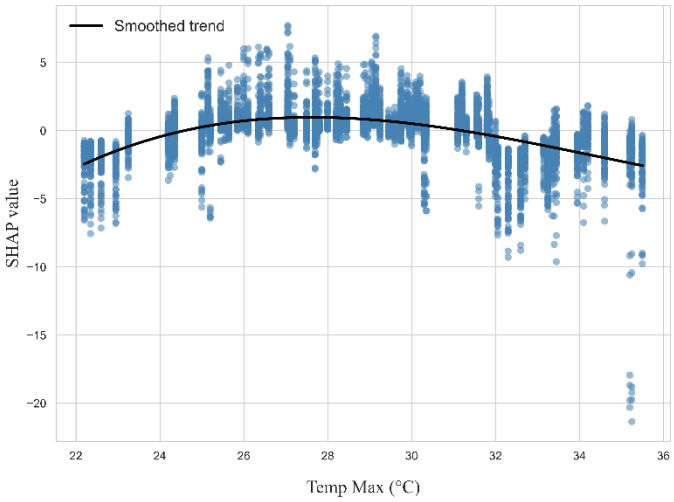
Fig. 8. SHAP dependence plots for station characteristics variables

4.3.2. Temporal and weather variables

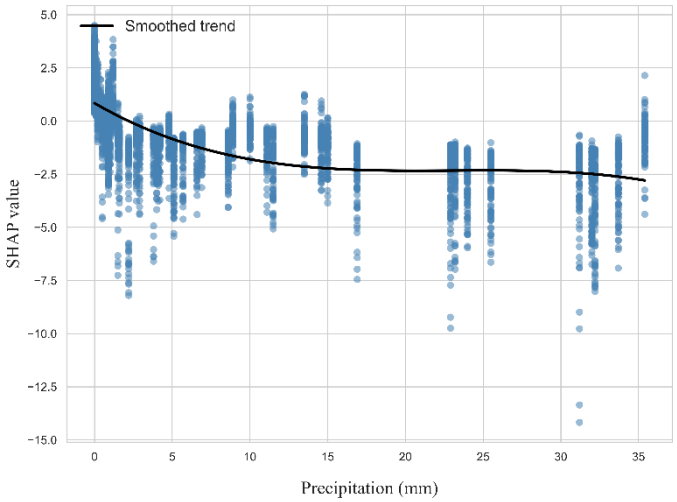
Figure 9 presents the temporal and weather variables. The weekend indicator (Fig. 9a) contributes positively, meaning that predicted demand per period is higher at weekends than on weekdays, which may reflect recreational and waterfront travel during the summer study period. Both peak-period indicators (Fig. 9e–f) have negative contributions, the AM peak more strongly. This should not be read as lower demand during peak hours. Demand was aggregated per period, and the off-peak reference covers more hours, whereas each peak period covers three (07:00–10:00 and 16:00–19:00). As a result, the mean demand per station, trip stage, and period is higher off-peak (39.83 trips) than in the PM peak (26.78) or the AM peak (12.16), which explains the negative contributions and their ordering. Per hour, however, demand is highest during the peaks (4.05 trips in the AM peak and 8.93 in the PM peak, compared with 2.21 off-peak), consistent with the hourly profile in Fig. 6. Weather conditions show clear nonlinear patterns. Maximum temperature (Fig. 9b) follows an inverted U-shape, with the highest contributions at about 26–29 °C before extreme heat reduces demand. Precipitation (Fig. 9c) shows a positive contribution on dry days that shifts into negative territory as soon as rain begins and then remains about constant. Maximum wind speed (Fig. 9d) has a declining contribution at higher speeds, and mean humidity (Fig. 9g) has little effect below about 75% and becomes increasingly negative at higher values.


Smoothed trend
SHAP value
Weekend

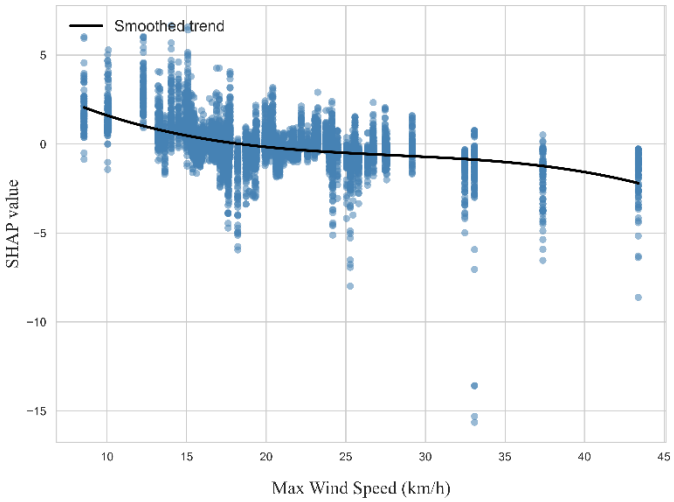
(a)


Smoothed trend
SHAP value
Temp Max (°C)

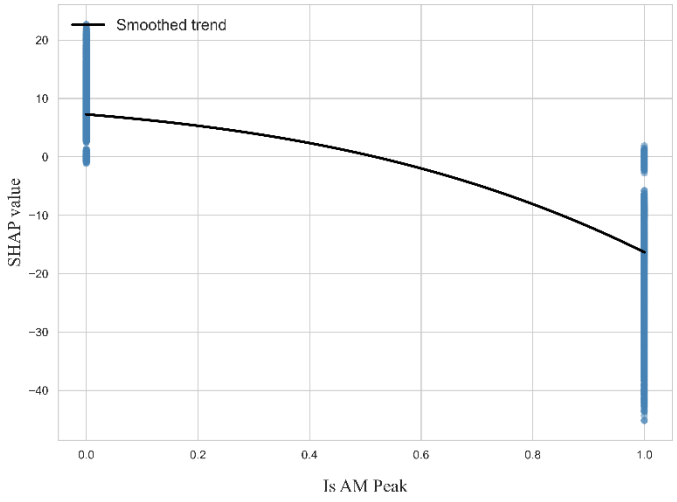
(b)


Smoothed trend
SHAP value
Precipitation (mm)

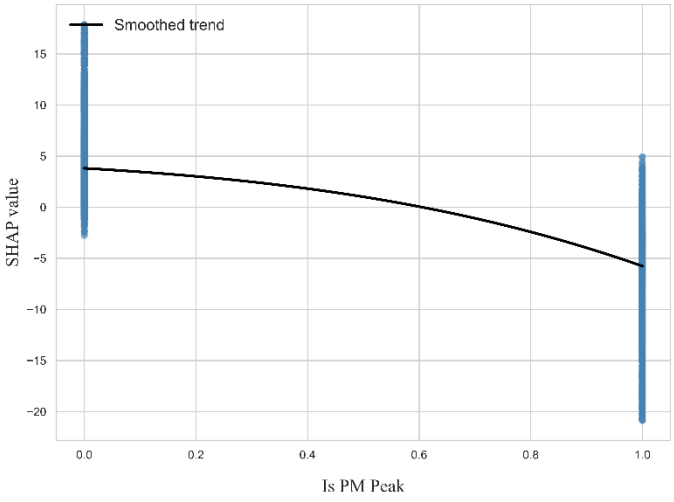
(c)


Smoothed trend
SHAP value
Max Wind Speed (km/h)

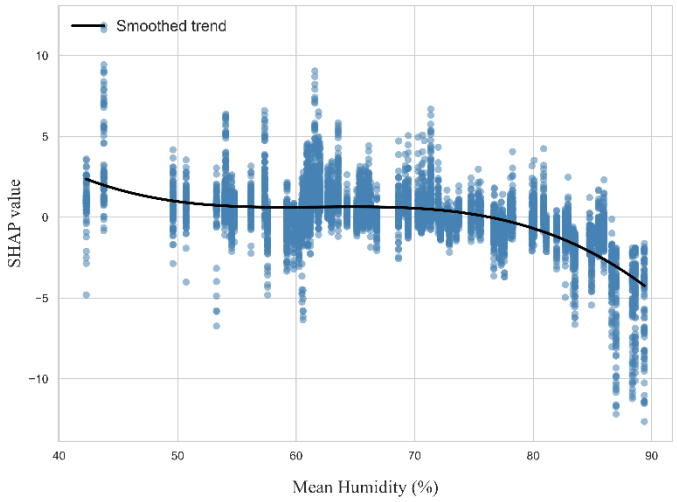
(d)


Smoothed trend
SHAP value
Is AM Peak

(e)


Smoothed trend
SHAP value
Is PM Peak

(f)

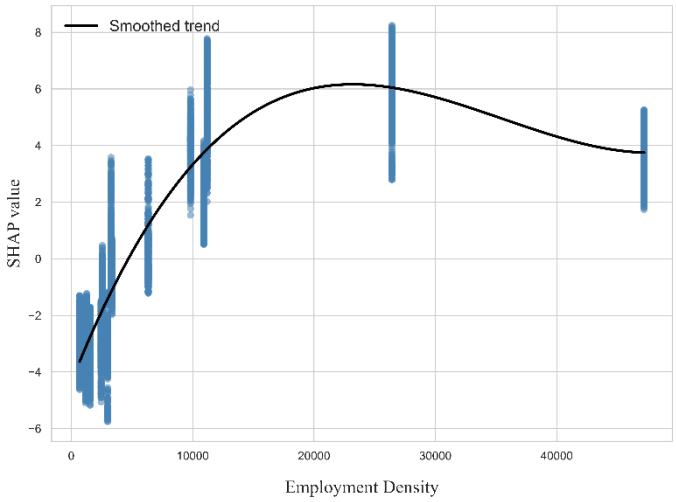


(g)

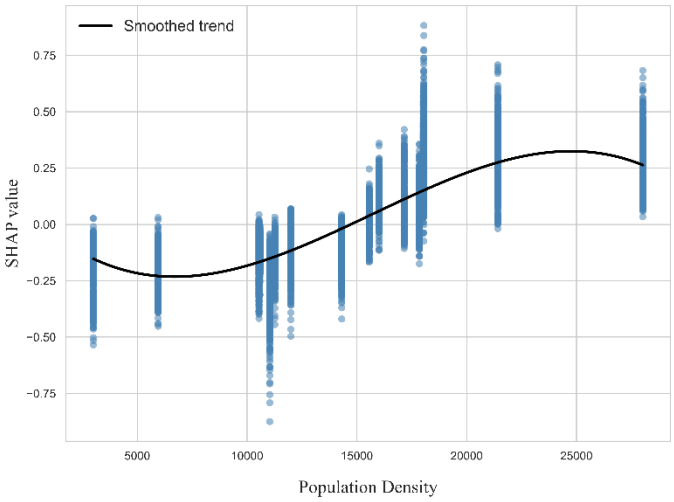
Fig. 9. SHAP dependence plots for Temporal and weather variables

4.3.3. Socio-demographic variables

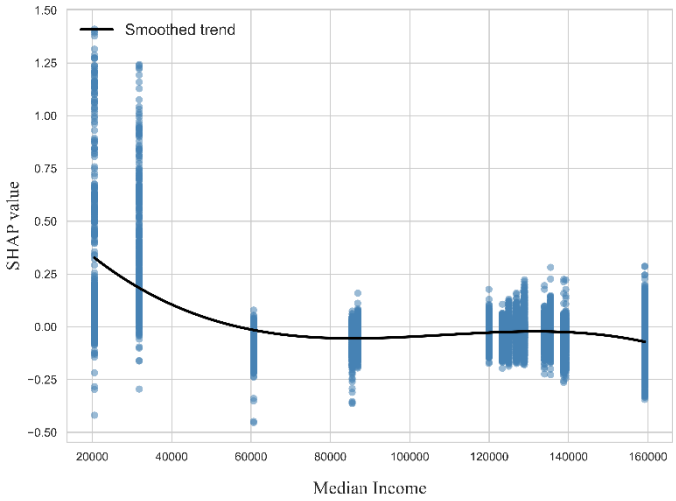
The dependence plots for socio-demographic variables (Fig. 10) show how catchment-level population and economic characteristics shape integrated bike–ferry demand. Employment Density (Fig. 10a) emerges as the predominant socio-demographic driver, showing a strong upward trajectory in which high concentrations of jobs significantly boost DBS-ferry demand. Population Density (Fig. 10b) displays an inverted U-shape, transitioning from negative contributions in lower-density areas to positive impacts that peak between ~20,000 and 25,000 residents per square kilometer. Household economic metrics reveal distinct non-linear dynamics: Median Income (Fig. 10c) shows its highest positive contribution in lower-income catchments (~$20,000–$30,000) before dropping into slightly negative territory across middle- and upper-income brackets, while Zero-Vehicle Share (Fig. 10d) exhibits an upward trend above, confirming strong transit reliance among car-free households. Educational Attainment (Fig. 10e) remains modest, shifting gradually from negative values in lower-education catchments to slight positive contributions above 50% higher education. Finally, age composition demonstrates distinct cohort dynamics: while the 18–24 age group (Fig. 10f) displays a subtle, near-neutral positive trend across higher concentrations, the 25–30 age group (Fig. 10g) exerts a much more pronounced influence, turning sharply positive once the cohort proportion exceeds ~25% and peaking above 40%. Overall, socio-demographic variables contributed slightly more than built-environment variables, although both categories were less influential than temporal and weather conditions and station characteristics.


Smoothed trend
SHAP value
Employment Density

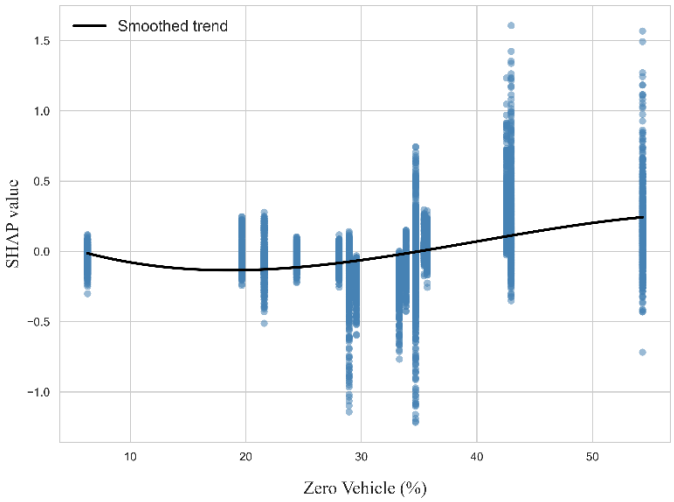
(a)


Smoothed trend
SHAP value
Population Density

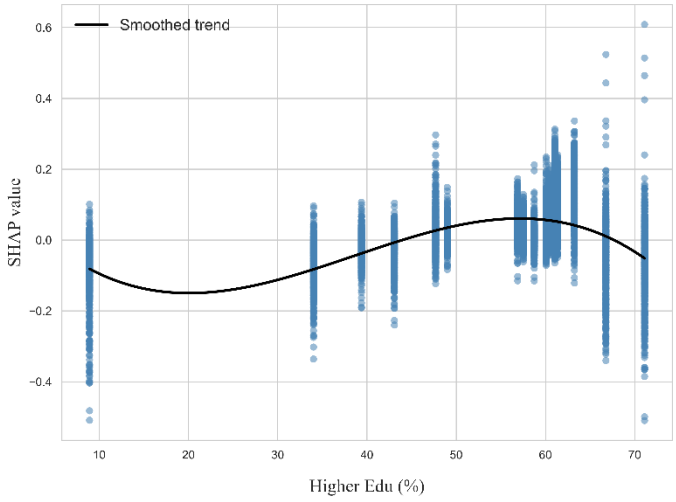
(b)


Smoothed trend
SHAP value
Median Income

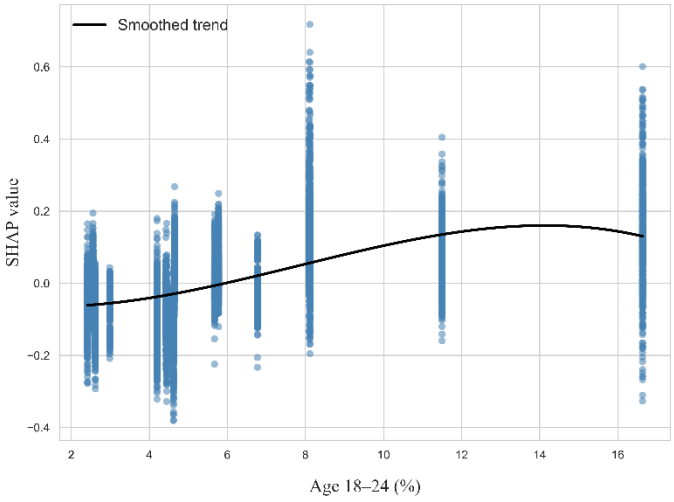
(c)


Smoothed trend
SHAP value
Zero Vehicle (%)

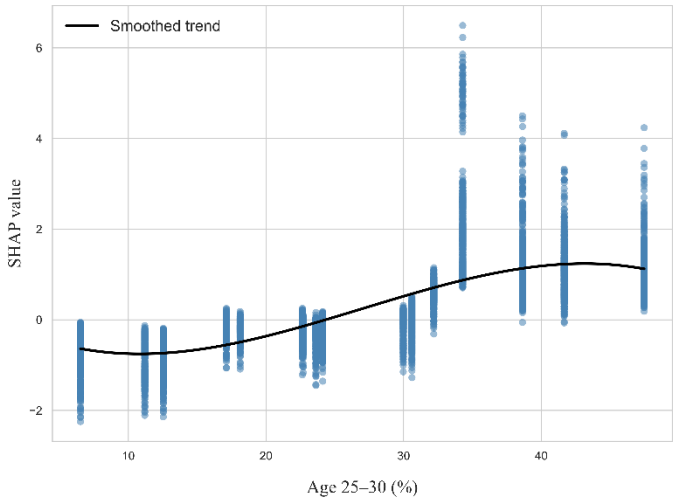
(d)


Smoothed trend
SHAP value
Higher Edu (%)

(e)


Smoothed trend
SHAP value
Age 18–24 (%)

(f)

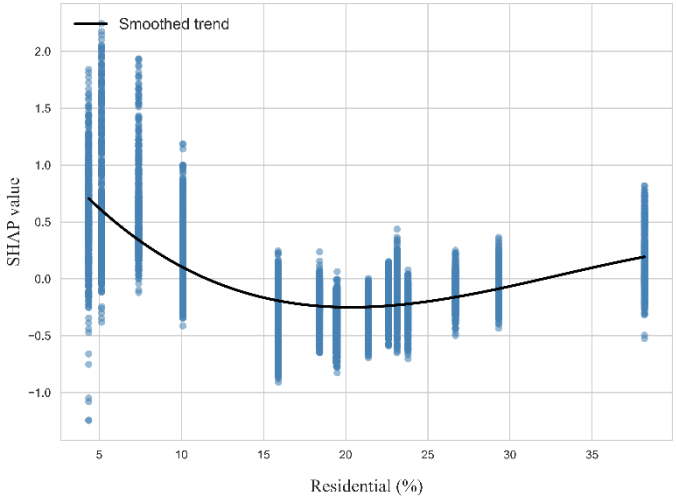


(g)

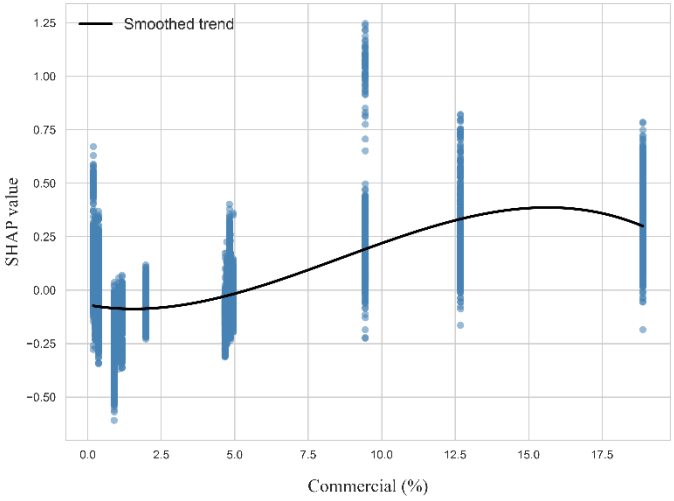
Fig. 10. SHAP dependence plots for Socio-demographic variables

4.3.4. Built environment variables

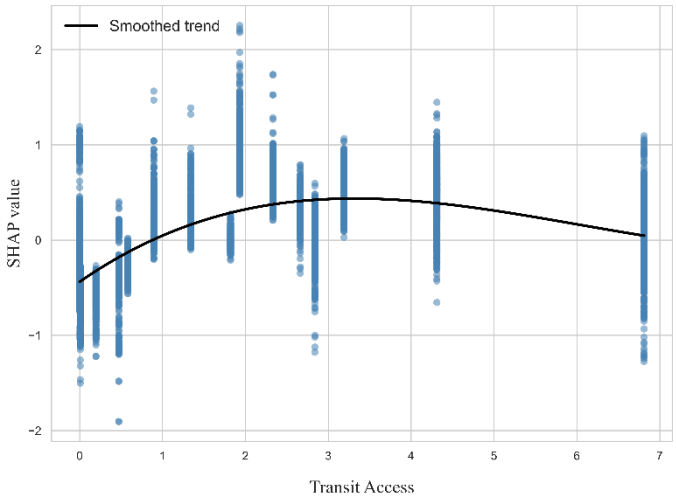
Figure 11 shows the built-environment variables. The residential share (Fig. 11a) has a U-shaped contribution, with negative values at moderate shares and positive values at the highest shares. The commercial share (Fig. 11b) shifts from negative contributions at low shares to positive ones at higher shares. Transit accessibility (Fig. 11c) moves steadily from negative to positive as accessibility increases. The industrial share (Fig. 11d) is neutral to negative over most of its range, whereas the institutional share (Fig. 11e) contributes strongly positively at about 8–13%. Intersection density (Fig. 11f) is positive above about 500, and total road length (Fig. 11g) shifts from negative values below about 40,000 m to stable positive values above. Land-use entropy (Fig. 11h) increases steadily, with positive contributions in mixed-use catchments (entropy above about 0.75).



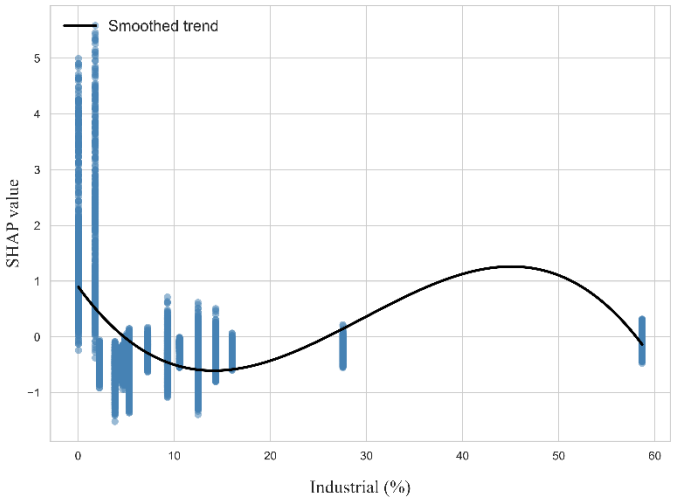
(a)



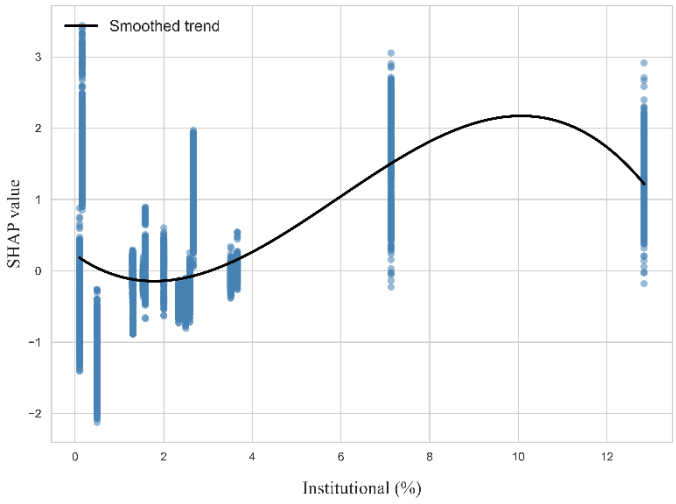
(b)



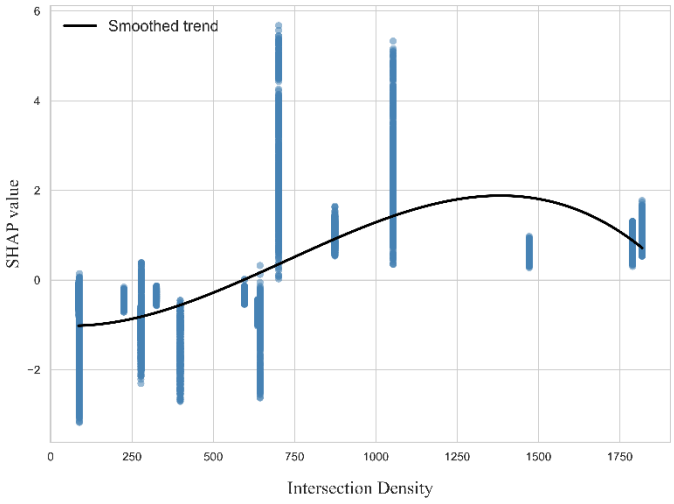
(c)



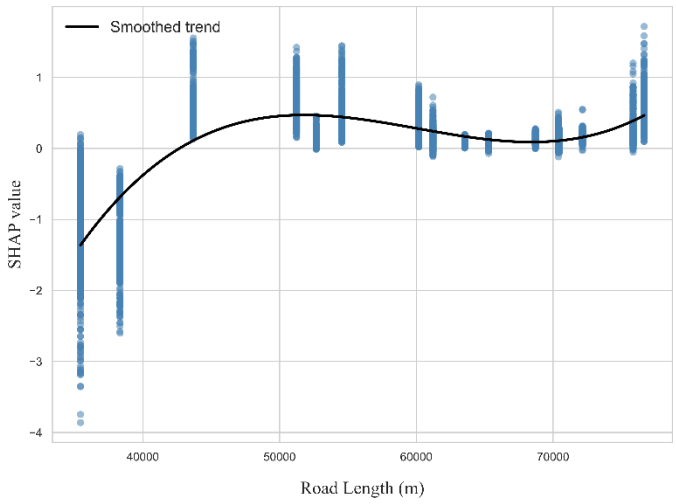
(d)



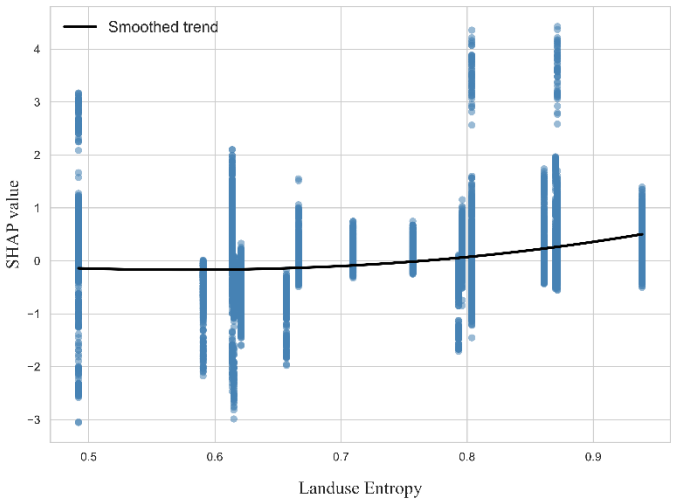
(e)



(f)



(g)



(h)

Fig. 11. SHAP dependence plots for Built environment variables

5. Discussion

Our findings provide a multidimensional understanding of the determinants shaping integrated docked bike-sharing (DBS)–ferry trips, revealing the relative importance of temporal and weather conditions, station characteristics, socio-demographic attributes, and the built environment. The results indicate that integrated travel demand is primarily associated with temporal patterns and station-level characteristics, while socio-demographic and built-environment variables provide additional insights into the spatial and population characteristics associated with this demand.

Station characteristics were the second most important feature category, accounting for 32.6% of total SHAP importance (Fig. 7b), and dock capacity and ferry distance were the two most important variables in this group (Fig. 7a). This finding highlights the importance of physical infrastructure and transfer accessibility in facilitating integration between bike-sharing and ferry services. Previous studies have emphasized the role of well-integrated bike-sharing systems in improving access to public transportation and supporting active travel (Gou, Feng et al. 2025, Ntamwiza and Bwire 2025, Alimo, Franceschini et al. 2026), and our results provide further evidence of the importance of station capacity and proximity in the DBS–ferry context. Dock capacity (Fig. 8a) exhibits a nonlinear relationship with predicted demand: its SHAP contribution is negative at capacities below approximately 40 docks and positive at larger stations, with the highest contributions at the stations with 76 and 123 docks. This pattern suggests that larger stations are associated with greater integrated trip demand, potentially because they accommodate more bikes and provide greater docking flexibility, which is consistent with the results of Zhang, Zheng et al. (2025) and Lu et al. (2026). Ferry proximity also exhibits a pronounced distance-decay pattern (Fig. 8b), with positive SHAP contributions at stations within approximately 210 m of the ferry stop and negative contributions at stations farther than approximately 225 m. This finding underscores the importance of minimizing the distance between bike-sharing stations and ferry terminals when designing convenient intermodal connections. Furthermore, trip stage (Fig. 8c) shows a modest difference: last-mile trips (Trip Stage = 0) exhibit a slightly more positive SHAP contribution to predicted demand than first-mile trips (Trip Stage = 1), although the two ranges overlap substantially. This pattern is consistent with previous studies reporting a greater prevalence of last-mile than first-mile bike–transit trips (Zhang, Meng et al. 2025, Lu, Li et al. 2026).

Temporal and weather conditions accounted for the largest share of SHAP importance (49.3%; Fig. 7b), demonstrating that integrated DBS–ferry demand varies with both travel timing

and meteorological conditions. The positive SHAP contribution associated with weekends (Fig. 9a) suggests that weekends are associated with higher predicted demand for integrated trips, potentially reflecting the influence of recreational activities, tourism, and waterfront travel in New York City during the summer study period. The negative SHAP contributions of the morning and evening peak-period indicators (Fig. 9e–f) should not be read as lower demand during peak hours. Because demand was aggregated into three periods of unequal length, with the peaks covering three hours each and the off-peak period eighteen hours, the mean number of trips per station, trip stage, and period is lower in the peaks (12.16 and 26.78) than off-peak (39.83). Per hour, the pattern reverses, with 4.05 and 8.93 trips in the morning and evening peaks compared with 2.21 off-peak, so integrated demand per hour is concentrated in the commuting peaks, particularly in the evening. Because the peak indicators partly reflect period length, we refitted the model on hourly rates (Appendix B). The combined importance of the two indicators was robust, whereas their individual order depended on how exposure time was handled, so the two indicators are best interpreted together. Weather variables further establish physical operational thresholds: Maximum Temperature (Fig. 9b) follows an inverted-U trend, maximizing positive utility around 26–30°C before extreme heat suppresses riding interest. Precipitation (Fig. 9c) exerts an immediate negative drag on rainfall onset, while Wind Speed (Fig. 9d) and Humidity (Fig. 9g) exert continuous, declining negative influences beyond muggy thresholds (~75% relative humidity). These non-linear climate constraints demonstrate that while baseline demand remains operable on dry, moderate weekdays, system utilization is highly vulnerable to adverse meteorological conditions, aligning with broader micromobility climate studies (Li, Chen et al. 2024, Xue, Zhang et al. 2024, Zhu, Wang et al. 2026).

Socio-demographic characteristics accounted for 9.1% of total SHAP importance (Fig. 7b), indicating a secondary but meaningful role in explaining variation in integrated trip demand. Among these variables, employment density was the most influential, with its SHAP contribution increasing substantially in areas with greater job concentrations (Fig. 10a). This relationship highlights the potential importance of employment destinations in shaping the spatial distribution of integrated DBS–ferry demand and is consistent with research identifying employment density as an important determinant of bike-sharing usage (Yang, Zhu et al. 2026) . Population density showed small contributions that were negative at lower values and positive at higher values (Fig. 10b). Household economic and demographic characteristics provide further insights into the areas associated with integrated travel. Higher zero-vehicle household shares were associated with slightly more positive SHAP contributions (Fig. 10d), suggesting that bike-sharing–ferry integration may be particularly relevant in locations characterized by greater reliance on non-car

transportation , although the effect is small. Median income exhibits a different pattern, with the strongest positive contributions at lower income levels, approximately $20,000–$30,000, followed by slightly negative contributions across middle- and higher-income ranges (Fig. 10c). Age composition also distinguishes between the two younger cohorts examined. Although the 18–24 age group shows a very modest contribution, the 25–30 age group becomes substantially more positive when its share exceeds approximately 25%, with the strongest contributions at shares above 40% (Fig. 10f–g). These findings suggest that integrated demand is associated with employment concentration and the presence of young adults in the 25–30 age group. However, because these characteristics describe station catchments rather than individual users, they should not be interpreted as direct evidence of the demographic composition or motivations of actual DBS–ferry passengers. The stronger contribution of the 25–30 age group nevertheless provides a useful basis for comparison with previous studies of shared bike usage in New York City, including Zhang et al. (2025), which identified a stronger association with the 18–24 cohort for shared e-bike use. This divergence suggests that the age profile associated with ferry-linked trips may differ from that of general shared e-bike use.

Built-environment characteristics accounted for 8.9% of total SHAP importance (Fig. 7b), indicating a smaller overall contribution than station characteristics and temporal and weather conditions. Nevertheless, several variables exhibited distinct nonlinear relationships with predicted demand. Commercial land-use share (Fig. 11b) showed increasingly positive SHAP contributions at higher values, suggesting that commercial areas may serve as important destinations for integrated travel, although its overall importance was low. Institutional land use (Fig. 11e) also showed strongly positive contributions when it accounted for approximately 8–12% of the catchment area, potentially reflecting the role of educational, civic, and other institutional destinations in generating travel demand. In contrast, residential land-use share (Fig. 11a) exhibited a U-shaped relationship, indicating that its association with demand varied across different levels of residential concentration. Land-use entropy (Fig. 11h) showed increasingly positive contributions at higher levels of land-use diversity, particularly above approximately 0.75, suggesting that mixed-use environments may be associated with greater integrated demand.

Street-network characteristics also played a role. Intersection density (Fig. 11f), measured as the number of street intersections per unit area within the 500 m buffer, showed increasingly positive SHAP contributions at higher values. Industrial land-use share (Fig. 11d) showed predominantly neutral to negative contributions across much of its range.

Overall, these findings suggest that the relationship between the built environment and integrated DBS–ferry demand depends on the combination of land-use composition, street-

network characteristics, and transit accessibility rather than on any single feature. Although their overall SHAP importance was comparatively modest, these characteristics provide additional insight into the urban contexts associated with integrated demand and complement the influence of station-level infrastructure. These findings are broadly consistent with previous studies examining the influence of the built environment on bike-sharing systems and public transportation (Bi, Li et al. 2022, Guo, Yang et al. 2022, Gou, Feng et al. 2025, Zhu, Dong et al. 2026).

5.1. Policy Recommendations

Bike-share operators and planners should consider prioritizing capacity management at ferry-adjacent stations, particularly where capacity may be insufficient relative to observed integrated demand. The association between dock capacity and predicted demand can inform station-level assessments, but it should be evaluated with local operating conditions before capacity is expanded. Given the importance of ferry proximity, planners should also consider the accessibility of connections between bike-share stations and ferry terminals, since all stations farther than about 225 m had negative contributions. Improving pedestrian access routes, reducing unnecessary walking distances, and providing clear wayfinding may facilitate transfers between the two modes. Because hourly demand is highest in the commuting peaks, bike availability and dock availability at ferry-adjacent stations are most critical in these periods. These recommendations should be read in the context of the study's scope. The analysis covers summer 2024, a period of high ridership for both systems (Citi Bike 2024, NYC Ferry 2024), which makes it a suitable period for studying integration but also means that the patterns may not hold in other seasons. Further validation with multiple seasons and years is needed before extending the findings to year-round planning.

5.2. Limitations of the Study

Several limitations should be considered. First, the analysis covers only summer 2024 and 16 ferry terminals, so the patterns may not represent year-round usage or other ferry networks. Second, because the station-level socio-demographic and built-environment variables take only one value per station, their dependence patterns rest on a small number of distinct catchments and should be interpreted cautiously. Third, integrated trips were identified from the temporal association between bike trips and ferry schedules within a 12-minute window, so they are potential rather than confirmed transfers, and longer transfers may be excluded. Appendix A shows that the feature-importance rankings are not sensitive to window length within the tested

range. Fourth, demand was aggregated into three periods of unequal length, which affects the interpretation of the peak-period indicators (Appendix B). Finally, SHAP values describe associations learned by the model and do not imply causal effects.

6. Conclusions

This study examined the determinants of integrated docked bike-sharing (DBS)–ferry demand in New York City using an XGBoost model benchmarked against OLS, a Poisson GLM, and LightGBM under a temporal holdout. The nonlinear models substantially outperformed the linear approaches, and XGBoost and LightGBM performed almost identically. XGBoost was therefore used for SHAP-based interpretation.

The SHAP analysis showed that temporal and weather conditions and station characteristics served as the primary drivers of integrated demand, with temporal and weather factors accounting for 49.3% of total feature importance and station characteristics accounting for 32.6%, followed by built-environment characteristics (8.9%) and socio-demographic attributes (9.1%) . Dock capacity and proximity to ferry terminals emerged as particularly influential factors, while last-mile connections exhibited slightly more positive SHAP contributions than first-mile connections, consistent with previous studies reporting a greater prevalence of last-mile bike–transit trips. Temporal patterns further revealed positive weekend contributions alongside negative contributions of the morning and evening peak indicators reflect the unequal length of the aggregation periods, since demand per hour is highest during the peaks. Meanwhile, commercial and institutional land uses, higher land-use diversity, intersection density, transit accessibility, and employment concentration provided additional insights into the urban contexts associated with stronger integrated demand. Furthermore, The larger contribution of the 25–30 age group than of the 18–24 group differs from the pattern reported for shared e-bike use in New York City (Zhang, Zheng et al. 2025), which suggests that the age profile associated with ferry-linked trips may differ from that of general e-bike use.

Overall, this study provides a data-driven understanding of the factors associated with bike-sharing and ferry integration and offers practical insights for improving station capacity, transfer accessibility, and coordinated multimodal planning. Future research should extend this analysis across multiple seasons, years, and ferry networks, and use confirmed transfers to assess the robustness and generalizability of the identified relationships.

Appendix A : Sensitivity Analysis of the Ferry-Transfer Window

To test whether the interpretation of the demand model depends on the 12-minute transfer window, integrated trips were re-identified with windows of 8, 10, 15, and 20 minutes using the same matching rules, and the model was refitted for each window. Windows shorter than 8 minutes were not considered, because the shortest scheduled travel time between ferry terminals is 8 minutes. All windows used the same panel (16 stations × 92 days × 3 periods × 2 trip stages; 8,832 observations), so the datasets differ only in the identified trips. Each model used the tuned hyperparameters, was trained on all days except the last 14, and was evaluated on those 14 days. Feature importance was the mean absolute SHAP value over all observations. Rankings were compared with the 12-minute baseline using Spearman's ρ and the overlap of the top-five features. In addition, the direction of each top-ten feature's effect was checked from the sign of the correlation between its value and its SHAP value. To separate window effects from model randomness, the baseline model was also refitted with five different random seeds.

The importance ranking was highly stable across windows (Table A1; Figure A1). Spearman's ρ with the baseline ranged from 0.942 to 0.977, compared with 0.975–0.993 across random seeds at 12 minutes. Windows from 8 to 15 minutes ($\rho = 0.973$–0.977) fall at or just below the lower end of the seed range, whereas the longest window (20 minutes) shows somewhat lower agreement ($\rho = 0.942$). The same six features, Is AM Peak, Dock Capacity, Ferry Distance, Is PM Peak, Employment Density, and Weekend, occupied the top six positions in every window, and Is AM Peak and Dock Capacity ranked first and second in all of them. Employment Density and Weekend exchanged fifth and sixth place at 15 and 20 minutes, which explains the top-5 overlap of 4 at those windows. An overlap of 4 also occurred in four of the five seed replications, so it is within model noise. None of the baseline's ten most important features changed its direction of effect in any window. Larger rank changes were limited to low-importance features with nearly tied SHAP values (for example, Intersection Density, ranked 7th to 17th, and Institutional share, 11th to 17th).

At the level of feature groups (Table A2), temporal and weather conditions remained the largest group in every window (47.0%–51.5%), followed by station characteristics (31.4%–32.9%), so these two groups together accounted for 78.4%–84.2% of the total importance. The share of temporal and weather variables increased slightly with the window (from 47.0% at 8 minutes to 51.5% at 20 minutes), whereas the shares of socio-demographic attributes (10.4% to 7.3%) and built-environment characteristics (11.1% to 8.5%) decreased. The ordering of these two smaller groups changed between windows, as their shares are similar.

Table A1. Model performance and agreement of the feature-importance ranking

Window (min)	Trips records	Holdout R^2	Spearman ρ	Top-5 overlap
8	177,530	0.845	0.973	5
10	206,748	0.852	0.977	5
12 (baseline)	231,902	0.855	1.000	5
15	265,517	0.866	0.976	4
20	310,278	0.862	0.942	4

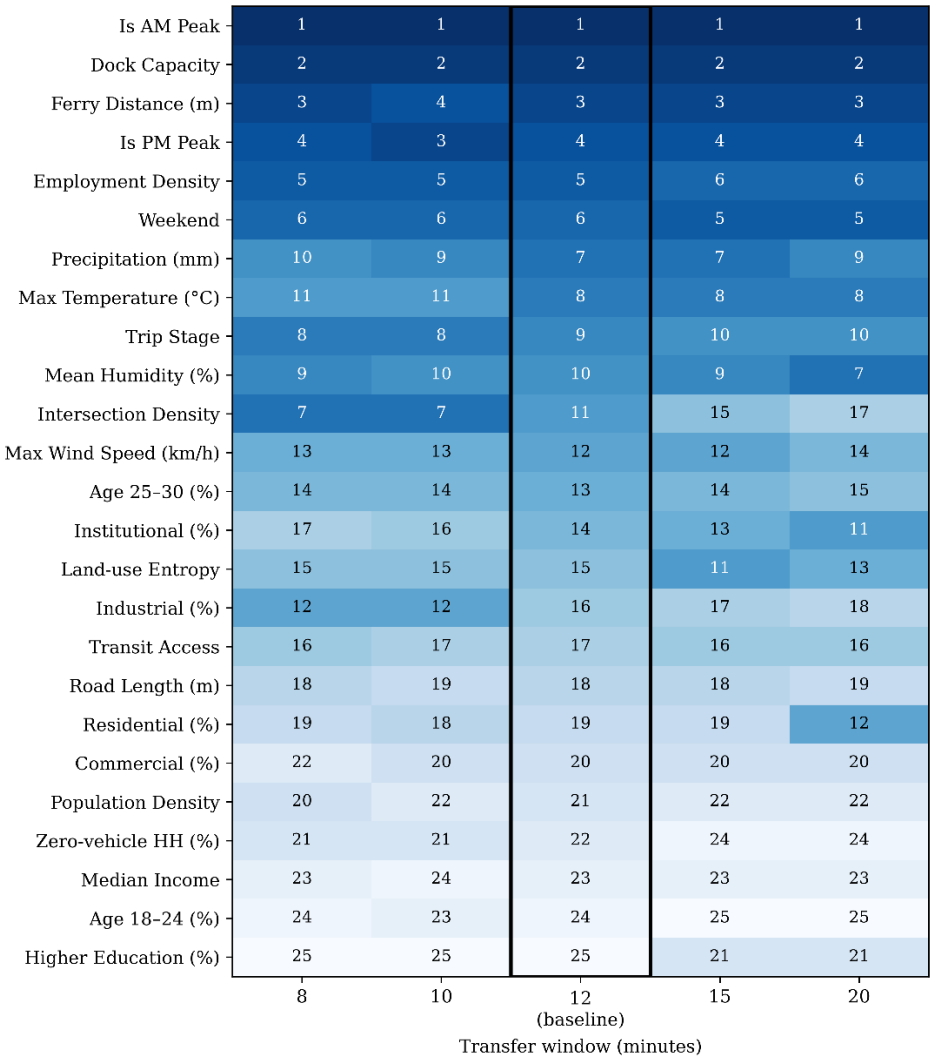


Fig. A1. Rank of each feature's mean absolute SHAP value for transfer windows of 8–20 minutes (1 = most important). The 12-minute baseline is outlined.

Table A2. GROUP IMPORTANCE SHARE

Window (min)	8 min	10 min	12 min	15 min	20 min
Station characteristics	31.4	31.5	32.59	32.9	32.7
Built environment	11.1	10.5	8.9	8.4	8.5
Socio-demographic	10.4	10.1	9.10	7.9	7.3
Temporal & weather	47.0	47.9	49.29	50.7	51.5

This analysis demonstrates that the feature-importance rankings are stable to the choice of transfer window within the tested range (8–20 minutes). It does not validate the integrated-trip identification heuristic: it cannot establish what proportion of the identified trips are genuine bike–ferry transfers rather than coincidental spatial–temporal co-occurrence, since the trip records contain no information on the traveller's linked journey. The identified integrated trips should therefore be read as potential, not confirmed, transfers

Appendix B : Sensitivity of the Importance Hierarchy to Period Length

The dependent variable is the number of integrated trips per station, date, period and trip stage, and the periods differ in length: 3 hours for each peak and 18 hours for off-peak, of which roughly 12 hours have ferry service. To test whether this exposure drives the importance hierarchy, the model was refitted on hourly trip rates (trips divided by period length), once unweighted and once weighted by period length, using the same panel, 25 predictors, hyperparameters and temporal holdout. Per hour, the evening peak is the busiest period (8.93 trips), followed by the morning peak (4.05) and off-peak (2.21, or about 3.3 per service hour), whereas the mean count per period is lowest in the morning peak (12.16) and highest off-peak (39.83).

Predictive performance on the count scale was similar across specifications (Table B1). The same five features were the most important in all three models, Dock Capacity ranked second in all of them, and the importance ranking was highly correlated with that of the count model (Spearman's $\rho = 0.962$ and 0.975, compared with 0.975–0.993 between reseeded count models). The group shares changed only modestly: temporal and weather variables remained the largest group (49.3% in the count model; 51.3% and 54.1% in the rate models), followed by station characteristics (32.6%; 27.6% and 27.4%), while socio-demographic and built-environment shares stayed at 9–12%. The main difference concerned the two period indicators: Is AM Peak ranked first in the count model but fourth and third in the rate models, whereas Is PM Peak moved from fourth to first. Their combined importance was similar (33.5% in the count model; 36.5% and 39.5% in the rate models). The period indicators therefore jointly dominate the temporal group in both specifications, but their individual order depends on how exposure time is handled, and they are interpreted together in the main text.

Table B1. Count and hourly-rate model comparison

Model	R^2 (counts)	Spearman ρ*	Rank: AM / PM peak
Count (main)	0.855	1.000	1 / 4
Hourly rate	0.838	0.962	4 / 1
Hourly rate, weighted	0.850	0.975	3 / 1

*Spearman's ρ measures the correlation between each model's feature-importance ranking and that of the main count model.

Code Availability

The Python code and analytical scripts developed for data preprocessing, machine learning model training, SHAP-based interpretation, and the sensitivity analyses reported in Appendices A and B are publicly available in the GitHub repository at https://github.com/amir-dehkordi/IntegratedBikeDemand.

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